

INTRODUCTION TO APPLIED NUMERICAL METHODS

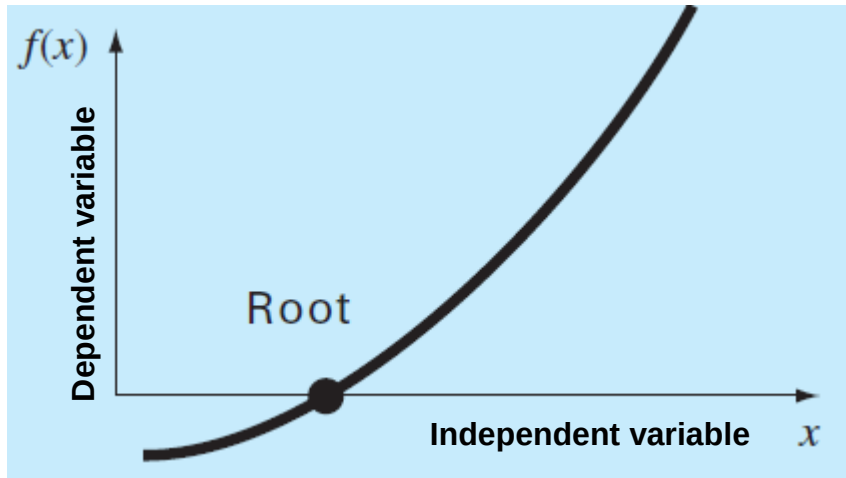
Overview

Lang Yuan, Associate Professor

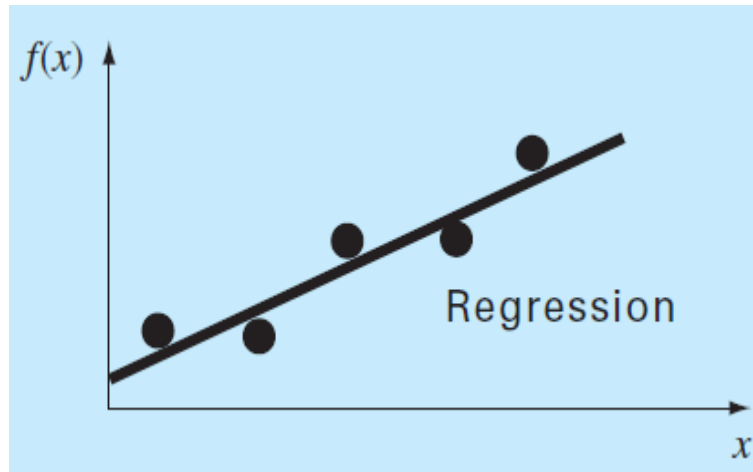


SUBJECT AREAS OF THIS COURSE

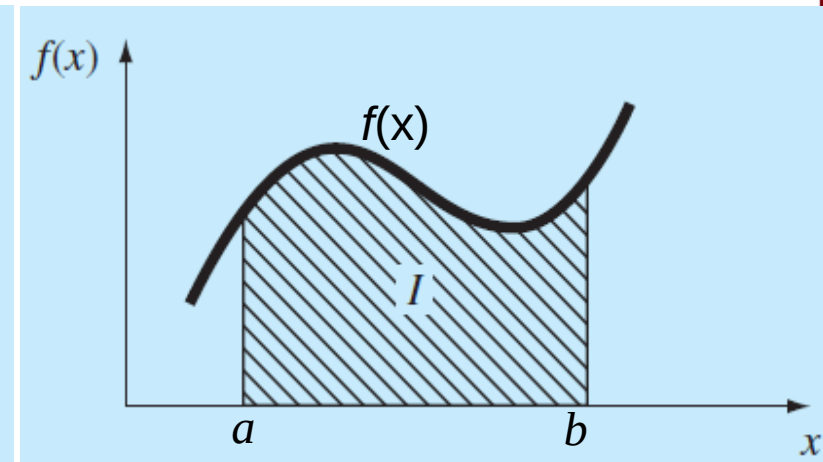
Roots of equations



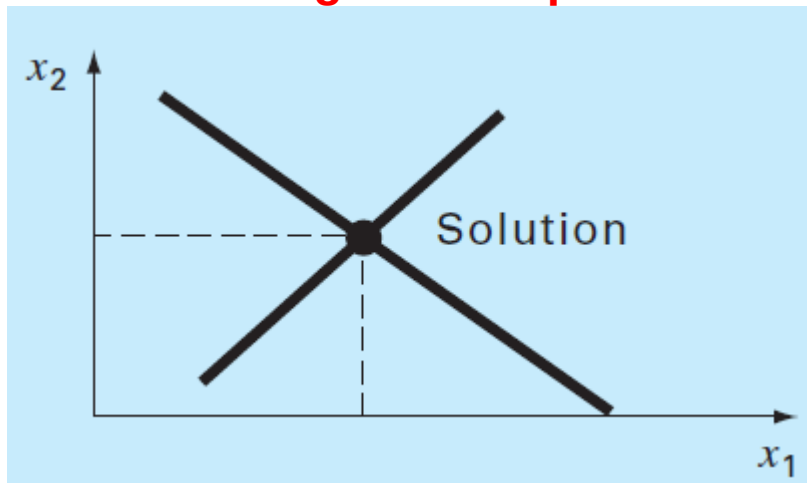
Regression



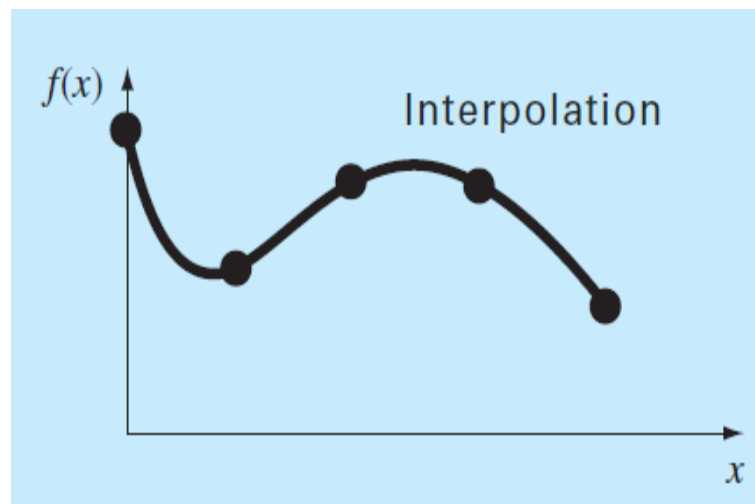
Integration



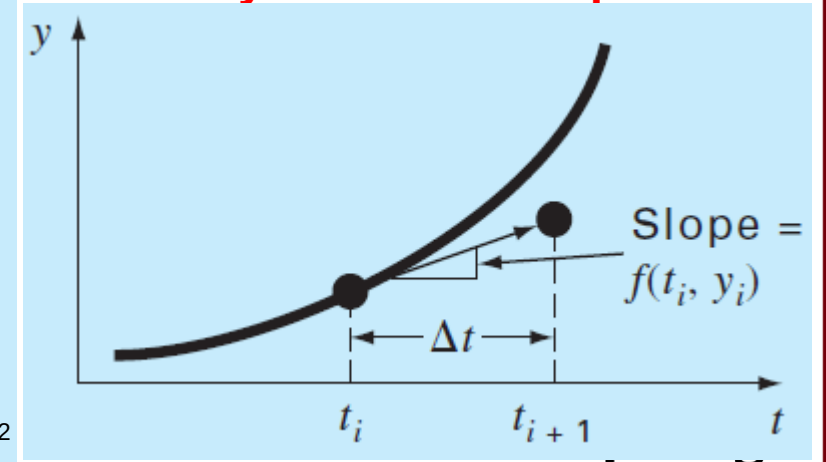
Linear Algebraic Equations



Interpolation



Ordinary Differential Equations



BASIS : SIGNIFICANT FIGURES AND ERRORS

- Significant figures
 - e.g. 0.0120
- Round-off error and truncation error
 - $1/3 = 0.333$
 - $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$
- True fractional relative error, approximated relative percent error
- Pre-specified percent error
 - Convention :

$$\varepsilon_s = (0.5 \times 10^{2-n})\%$$

the result is correct to *at least* n significant figures

TAYLOR SERIES

$$f(x_{i+1}) = f(x_i) + f'(x_i)(x_{i+1} - x_i) + \frac{f''(x_i)}{2!}(x_{i+1} - x_i)^2 + \frac{f^{(3)}(x_i)}{3!}(x_{i+1} - x_i)^3 + \dots + \frac{f^{(n)}(x_i)}{n!}(x_{i+1} - x_i)^n + R_n$$

Remainder

Step size: $h = x_{i+1} - x_i$

Query
point

Expansion
point

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

Example cases:

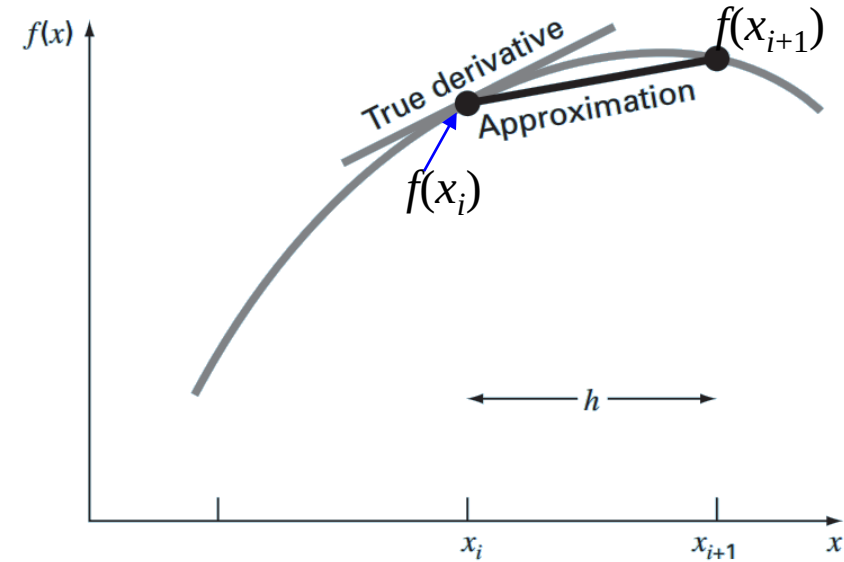
$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

NUMERICAL DIFFERENTIATION

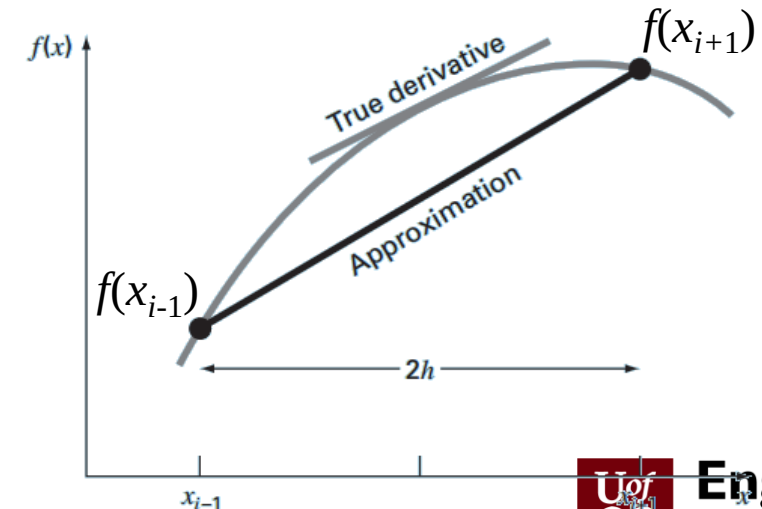
- Forward difference

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - O(h)$$



- Backward difference

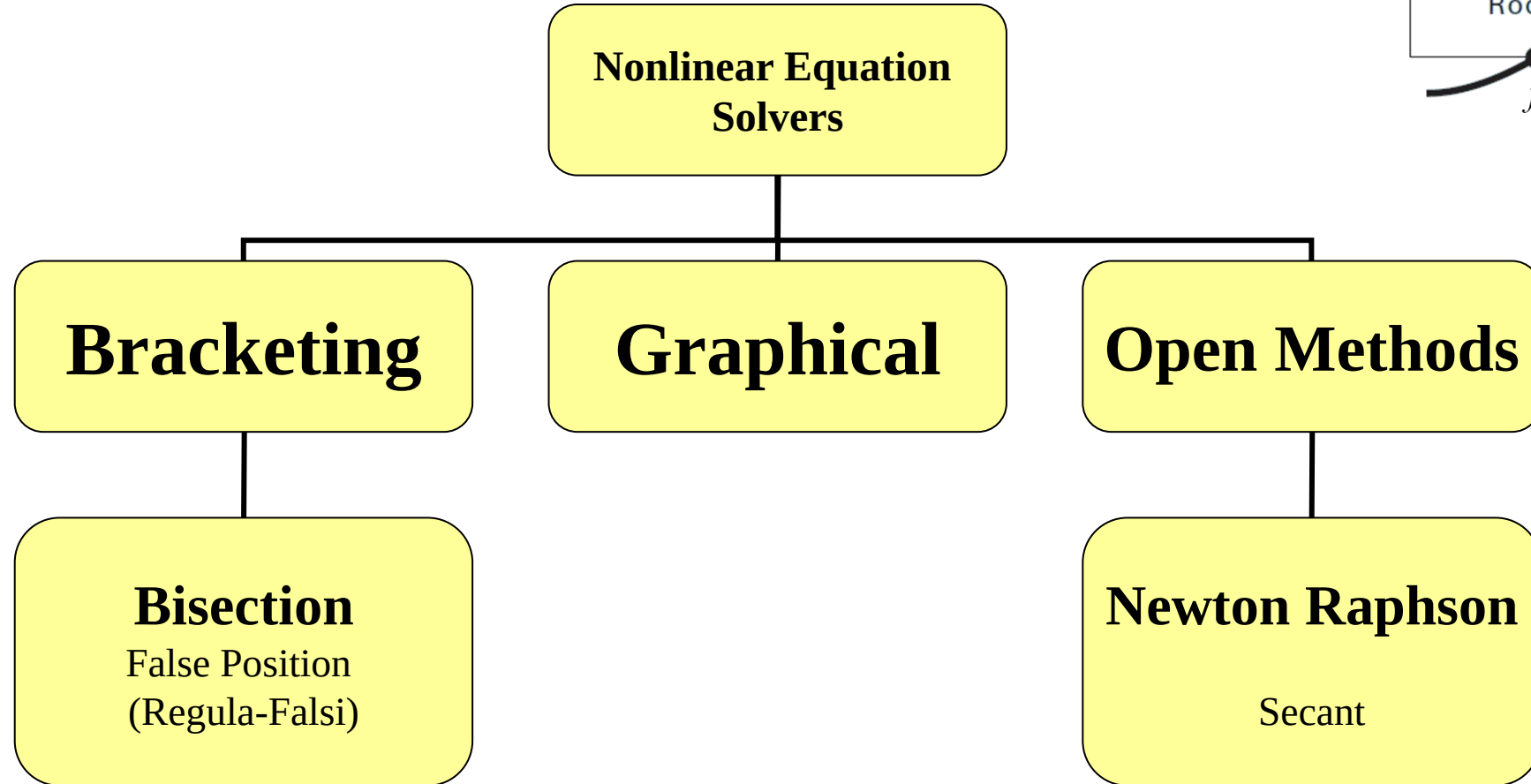
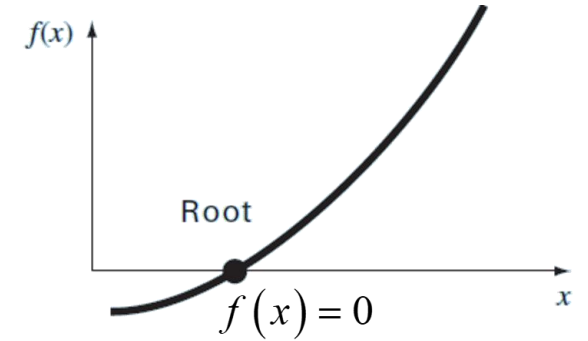
$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$



- Centered difference

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} - O(h^2)$$

OUTLINE OF APPROACHES



BISECTION METHOD: THE BISECTION PROCESS

Step 1: Choose lower and upper bounds of x_l and x_u for the root. Ensure the functional value changes sign over the interval, i.e., checking and ensuring

$$f(x_l) f(x_u) < 0$$

Step 2: Estimate the root x_r by setting

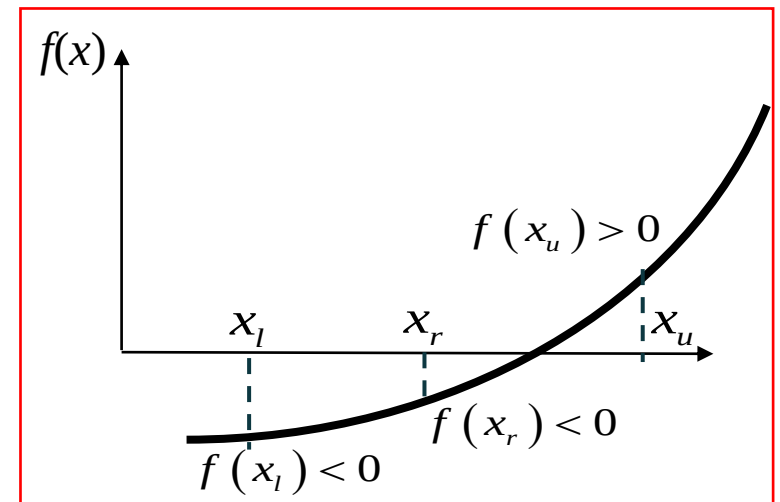
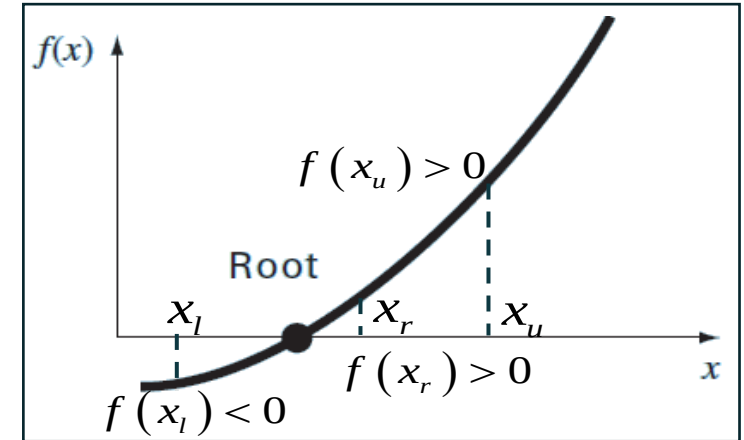
$$x_r = \frac{x_l + x_u}{2}$$

Step 3: Search the subinterval according to sign changes to determine the root location:

(3.1) If $f(x_l)f(x_r) < 0$, the root lies in the lower subinterval, i.e., $x_l < x < x_r$. Therefore, set $x_u = x_r$ and return to step 2.

(3.2) If $f(x_l)f(x_r) > 0$, the root lies in the upper subinterval, i.e., $x_r < x < x_u$. Therefore, set $x_l = x_r$ and return to step 2.

(c) If $f(x_l)f(x_r) = 0$, the root equals x_r and terminate the computation.



THE NEWTON-RAPHSON METHOD

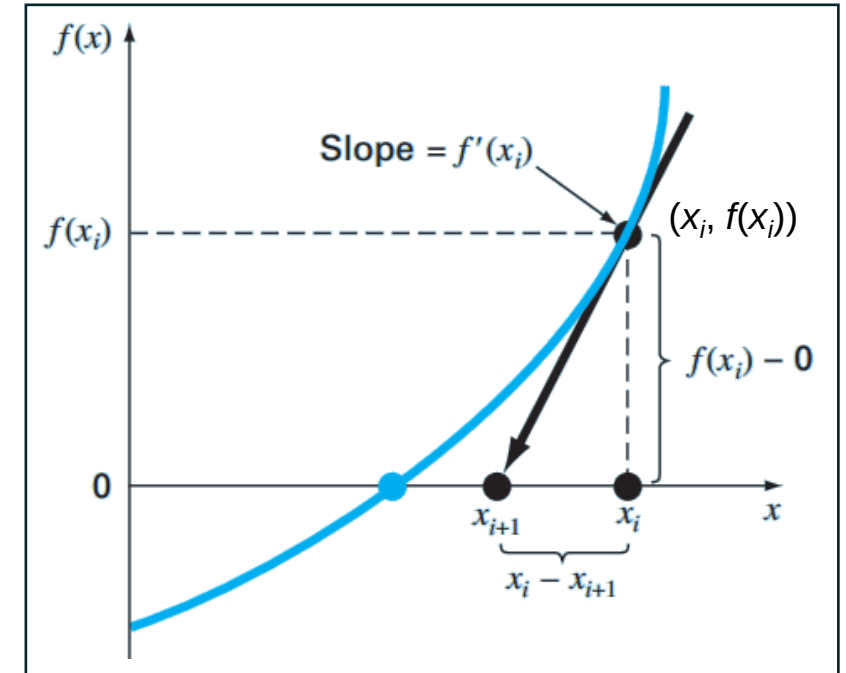
Principle

- Open method
- Most widely used root-finding method
- Use tangent to obtain an improved estimate for the true root

Approximation of the First-order Derivative using Forward Difference

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}} \Rightarrow x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Newton-Raphson Method



Numerical Methods for Engineers, 7th Ed.
(S.C.Chapra and R.P.Canale)

LINEAR EQUATIONS: MATRIX NOTATION

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nm} \end{bmatrix}$$

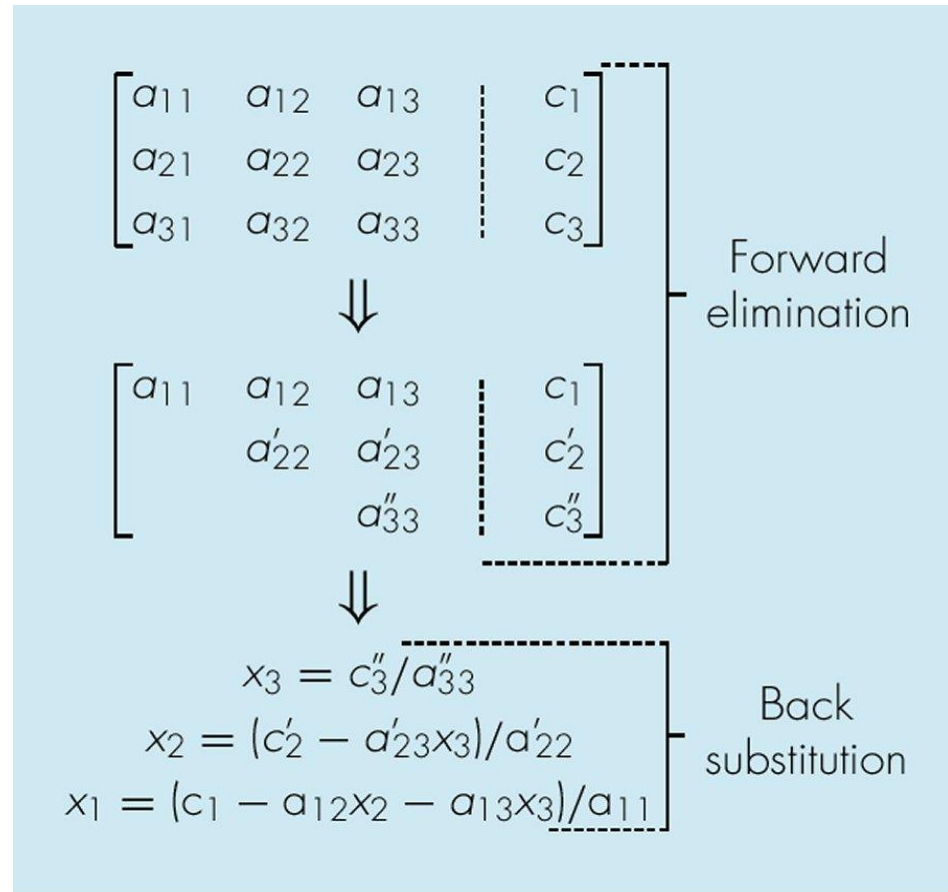
Column 3

Row 2

- Special types of square matrices: symmetric, diagonal, identity, upper triangular, lower triangular
- Matrix operations: addition, subtraction, multiplication, transpose, augmentation

GAUSS ELIMINATION

- Forward elimination of unknowns
- Back substitution



GAUSS ELIMINATION

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

- General process for a 3 by 3 matrix
- Forward elimination
 - Keep the first equation and reconstruct the second and until the last one
 - Keep the second equation and reconstruct the third one until the last one
 -
 - Elimination :
 - $(2)^1 = (2) - (1) * \frac{a_{21}}{a_{11}} \rightarrow \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}}\right)x_2 + \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}}\right)x_3 = b_2 - \frac{b_1a_{21}}{a_{11}}$
 - $(3)^1 = (3) - (1) * \frac{a_{31}}{a_{11}} \rightarrow \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}}\right)x_2 + \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}}\right)x_3 = b_3 - \frac{b_1a_{31}}{a_{11}}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}}\right) & \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}}\right) \\ 0 & \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}}\right) & \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}}\right) \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 - \frac{b_1a_{21}}{a_{11}} \\ b_3 - \frac{b_1a_{31}}{a_{11}} \end{Bmatrix}$$

GAUSS ELIMINATION

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}^1 & a_{23}^1 \\ 0 & a_{32}^1 & a_{33}^1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2^1 \\ b_3^1 \end{Bmatrix} \quad \begin{matrix} (1) \\ (2)^1 \\ (3)^1 \end{matrix}$$

$$a_{22}^1 = \left(a_{22} - \frac{a_{12}a_{21}}{a_{11}} \right)$$

$$a_{23}^1 = \left(a_{23} - \frac{a_{13}a_{21}}{a_{11}} \right)$$

$$a_{32}^1 = \left(a_{32} - \frac{a_{12}a_{31}}{a_{11}} \right)$$

$$a_{33}^1 = \left(a_{33} - \frac{a_{13}a_{31}}{a_{11}} \right)$$

$$b_2^1 = b_2 - \frac{b_1a_{21}}{a_{11}}$$

$$b_3^1 = b_3 - \frac{b_1a_{31}}{a_{11}}$$

- Forward elimination

- $(3)^2 = (3)^1 - (2)^1 * \frac{a_{23}^1}{a_{22}^1} \rightarrow \left(a_{33}^1 - \frac{a_{23}^1a_{32}^1}{a_{22}^1} \right) x_3 = b_3^1 - \frac{b_2^1a_{32}^1}{a_{22}^1}$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}^1 & a_{23}^1 \\ 0 & 0 & a_{33}^2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2^1 \\ b_3^2 \end{Bmatrix}$$

$$a_{33}^2 = \left(a_{33}^1 - \frac{a_{23}^1a_{32}^1}{a_{22}^1} \right)$$

$$b_3^2 = b_3^1 - \frac{b_2^1a_{32}^1}{a_{22}^1}$$

- Get the solution: Back substitution

- $x_3 \rightarrow x_2 \rightarrow x_1$

LU DECOMPOSITION

Process

- **Step 1:** LU Decomposition: $[A]$ is factorized or “decomposed” into lower $[L]$ and upper $[U]$ triangular matrices, i.e., $[A] = [L][U]$
- **Step 2:** Substitution step: (i)

$$[L][U]\underbrace{\{x\}}_{\{D\}} = \{B\}$$

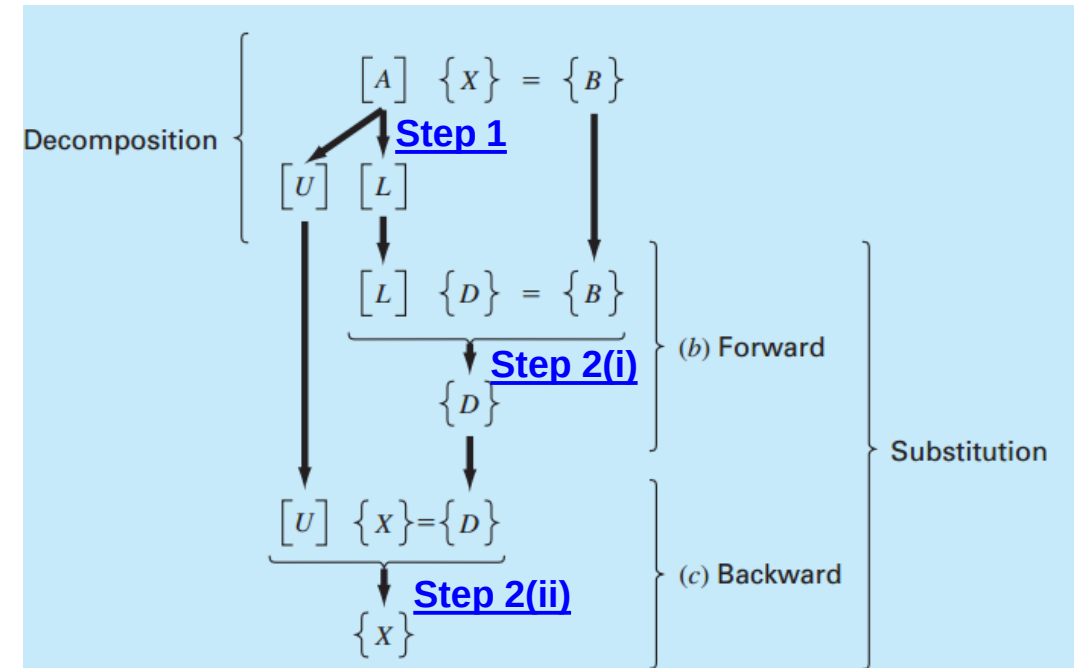
Introduce an intermediate vector $\{D\}$

$$[L]\{D\} = \{B\}$$

Forward Substitution

$$\{D\}$$

Backward Substitution (ii) $[U]\{X\} = \{D\}$



LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step

Step 1: Forward Elimination

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

$$\downarrow [A]: \text{Row}(2) - \text{Row}(1) \times \left(\frac{a_{21}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Define: } f_{21} = \left(\frac{a_{21}}{a_{11}} \right)$$

$$\downarrow [A]: \text{Row}(3) - \text{Row}(1) \times \left(\frac{a_{31}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\text{Define: } f_{31} = \left(\frac{a_{31}}{a_{11}} \right)$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step



Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$



$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

$$[A]: \text{Row}(3') - \text{Row}(2') \times \left(\frac{a'_{32}}{a'_{22}} \right)$$

Define: $f_{32} = \left(\frac{a'_{32}}{a'_{22}} \right)$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Key idea: use f_{21} , f_{31} , f_{32} , ... to assemble the lower triangular matrix $[L]$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$$[U] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \quad \rightarrow \quad [A] \rightarrow [L][U]$$

$$\begin{aligned} l_{11} &= 1 & l_{12} &= 0 & l_{13} &= 0 \\ l_{21} &= f_{21} & l_{22} &= 1 & l_{23} &= 0 \\ l_{31} &= f_{31} & l_{32} &= f_{32} & l_{33} &= 1 \end{aligned}$$

$$\begin{aligned} u_{11} &= a_{11} & u_{12} &= a_{12} & u_{13} &= a_{13} \\ u_{21} &= 0 & u_{22} &= a'_{22} & u_{23} &= a'_{23} \\ u_{31} &= 0 & u_{32} &= 0 & u_{33} &= a''_{33} \end{aligned}$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Process

- **Step 2(i):** Forward substitution step to obtain {D}

$$[L]\{D\} = \{B\}$$

$$d_i = b_i - \sum_{j=1}^{i-1} l_{ij} d_j \quad \text{for } i = 2, 3, \dots, n$$

$$\begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

- **Step 2(ii):** Back-substitution step to obtain {x}

$$[U]\{X\} - \{D\} = 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix}$$

$$x_n = \frac{d_n}{u_{nn}}$$

$$x_i = \frac{d_i - \sum_{j=i+1}^n u_{ij} d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \dots, 1$$

MATRIX INVERSION

- Inversion of $[A]$: $[A][A]^{-1} = [I]$
- The inversion can be calculated in a column-by-column fashion by generating solutions with unit vectors as the right-hand-side constants.
- Example:
 - $[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
 - $[A]^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$
 - $[A][A]^{-1} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 - Calculated the matrix in a column-by-column fashion (solving two linear equations):
 - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

CURVE FITTING

Curve Fitting Approaches

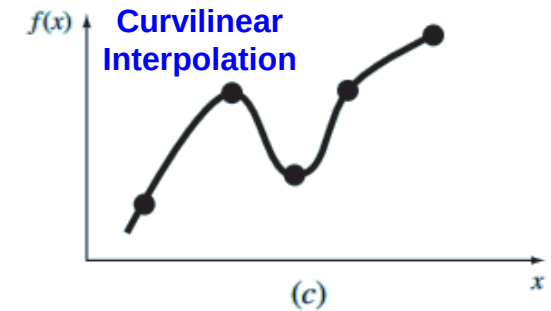
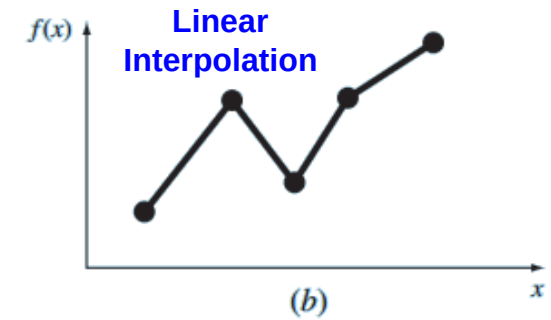
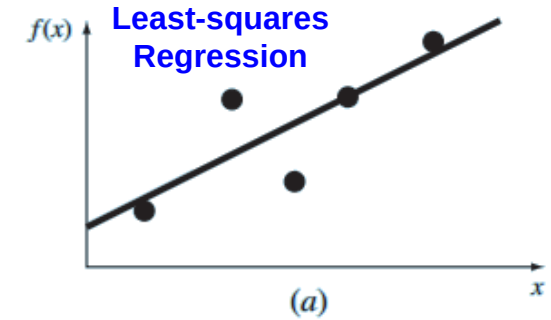
- Regression
 - Least Squares Regression
- Interpolation
 - Data is very precise

Statistics Measures

- Mean
- Standard deviation

$$s_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}}$$

- Variance s_y^2
- Coefficient of Variance (C.V), $\frac{s_y}{y_{mean}}$



LEAST-SQUARES FIT OF A STRAIGHT LINE

Linear regression: $y = a_0 + a_1x$

Sum of the Squares of the Residual

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^n (y_i - a_0 - a_1x_i)^2$$

Normal
Equations

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$



$$a_0 = \bar{y} - a_1 \bar{x}$$

Mean
of y

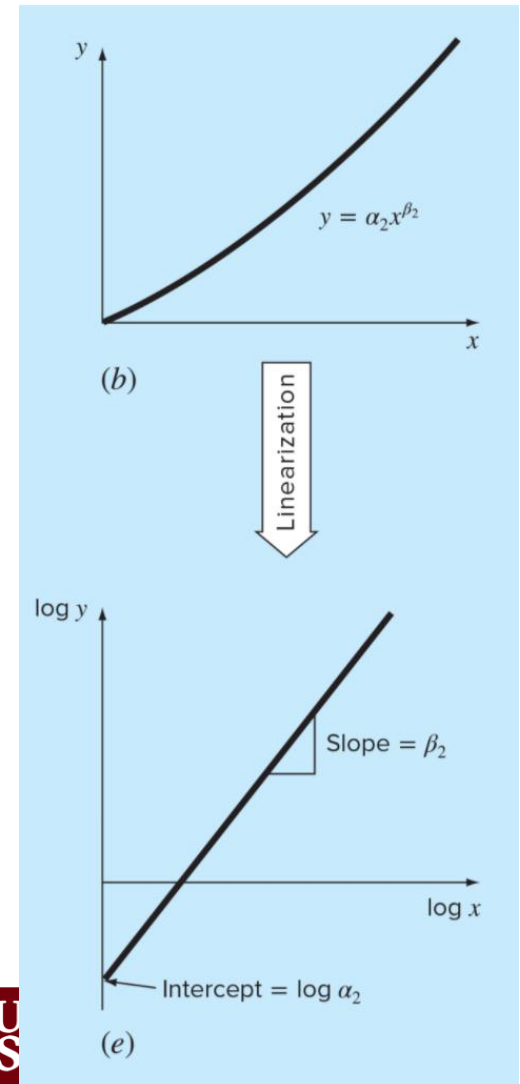
Mean
of x

For Linear Regression: we need

$$\left\{ \begin{array}{l} \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n x_i^2 \\ \sum_{i=1}^n x_i \quad \text{or} \quad \bar{x} \\ \sum_{i=1}^n y_i \quad \text{or} \quad \bar{y} \end{array} \right.$$

LINEARIZATION OF NONLINEAR RELATIONSHIPS

- Nonlinear model for simple power equation
 - $y = ax^b$
- Transform the equation to linear expression : e.g. take $\ln$ at both sides
 - $\ln(y) = \ln(ax^b) = \ln(a) + b \ln(x)$
 - Linear equation for $\ln(y) \sim \ln(x)$
- $y = ae^{bx} \rightarrow \ln(y) = \ln(a) + bx$
- $y = \frac{ax}{b+x} \rightarrow \frac{1}{y} = \frac{b+x}{ax} = \frac{1}{a} + \frac{b}{a} \frac{1}{x}$



ERROR OF LINEAR REGRESSION

Sum of the Squared Error

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

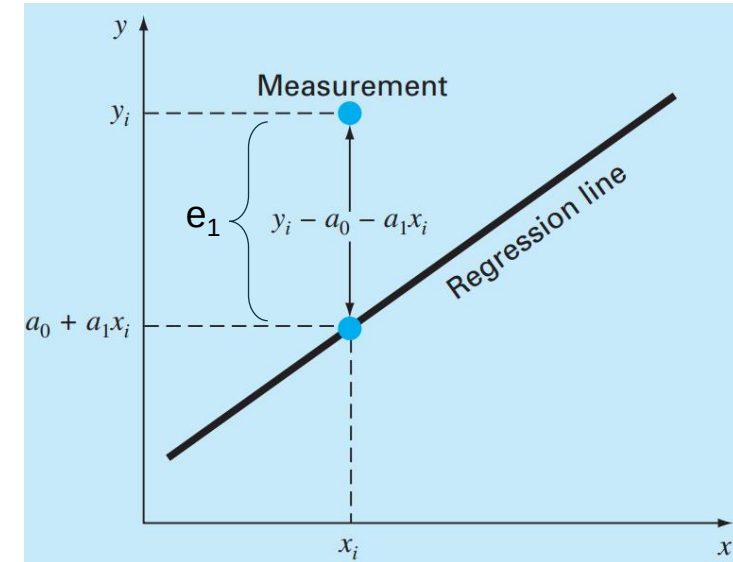
Standard Error of the Estimate ($s_{y/x}$)

$$s_{y/x} = \sqrt{\frac{S_r}{n-2}}$$

← Degree of freedom (for data used in linear regression)

$s_{y/x}$: Quantifies the spread around the regression line; **but it does not provide a quantitative and relative measure of “goodness of fit”**

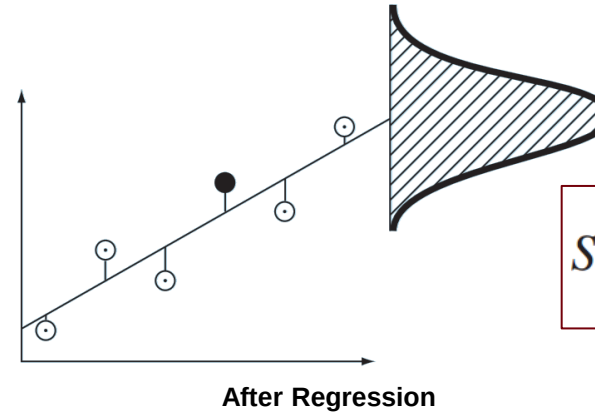
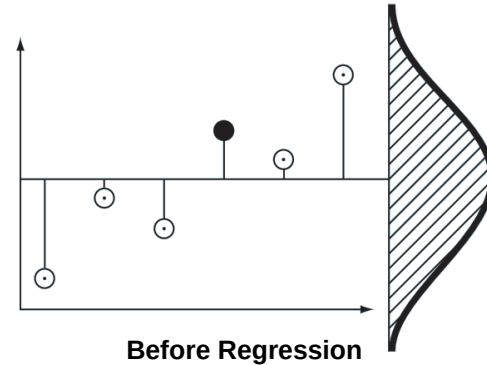
y/x notation designates the error is for a predicted value of y corresponding to a particular x



ERROR OF LINEAR REGRESSION

Coefficient of Determination

$$S_t = \sum (y_i - \bar{y})^2$$



$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

$$r^2 = \frac{S_t - S_r}{S_t}$$

Improvement or error reduction
due to the linear regression

Error before the linear regression

The coefficient of
determination

r : correlation coefficient $r = \sqrt{\frac{S_t - S_r}{S_t}}$

$S_r=0$ and $r = r^2 = 1$: the line explains 100% of the variability of the data (perfect fit)

$S_r = S_t$ and $r = r^2 = 0$: regression yields no improvement (poor fit)

LEAST-SQUARES REGRESSION OF A 2ND ORDER POLYNOMIAL

2nd order polynomial regression: $y = a_0 + a_1x + a_2x^2$

Three Equations and Three unknowns: a_0 , a_1 , and a_2

$$\begin{cases} (n)a_0 + (\sum x_i)a_1 + (\sum x_i^2)a_2 = \sum y_i \\ (\sum x_i)a_0 + (\sum x_i^2)a_1 + (\sum x_i^3)a_2 = \sum x_i y_i \\ (\sum x_i^2)a_0 + (\sum x_i^3)a_1 + (\sum x_i^4)a_2 = \sum x_i^2 y_i \end{cases}$$

- Standard Error

$$S_r = \sum_{i=1}^n (y_i - a_0 - a_1x_i - a_2x_i^2)^2$$

$$s_{y/x} = \sqrt{\frac{S_r}{n-3}}$$

For 2nd Order Polynomial Regression: we need

$$\left\{ \begin{array}{l} n \\ \sum x_i y_i \quad \sum x_i^2 y_i \\ \sum x_i^2 \quad \sum x_i^3 \quad \sum x_i^4 \\ \sum x_i \quad \text{or} \quad \bar{x} \\ \sum y_i \quad \text{or} \quad \bar{y} \end{array} \right.$$

LINEAR INTERPOLATION

Linear Interpolation

- The interpolated value lies on a straight line connecting two data points $(x_0, f(x_0)), (x_1, f(x_1))$

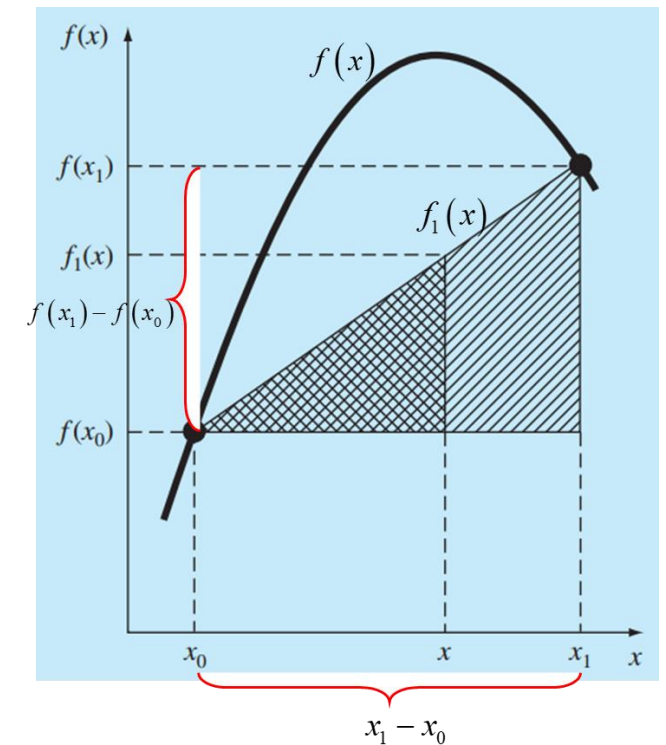
Interpolated value @ x

$$\frac{f_1(x) - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Subscript "1": First-order (Linear) interpolating polynomial for " f "

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$$

Finite-divided-difference approximation
of the first derivative



The smaller the interval between data points, the better the approximation!

QUADRATIC INTERPOLATION

Quadratic Interpolation

- Introduce some curvature into the line connecting the points
- A 2nd-order polynomial (quadratic polynomial): three data points are available

Subscript "2": 2nd-order (quadratic)
interpolating polynomial for "f"

2nd order curvature

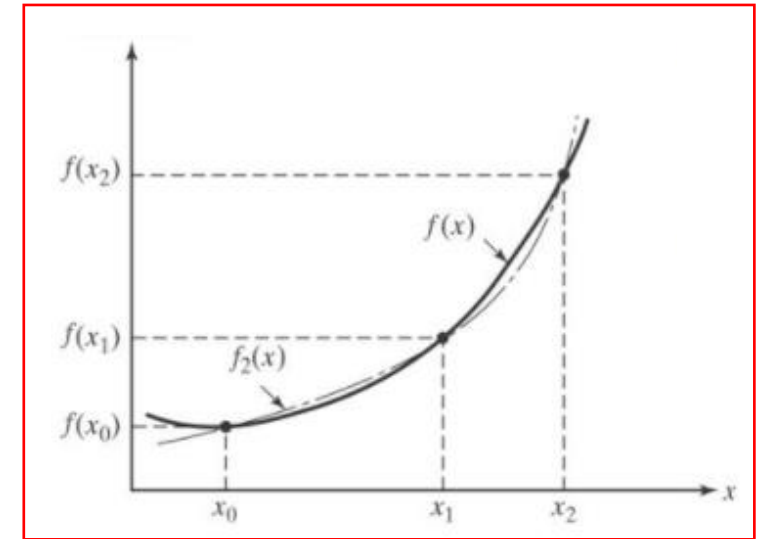
$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$@ x = x_0, b_0 = f(x_0)$$

$$@ x = x_1, b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Approximated first derivative/Slope

$$@ x = x_2, b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$



NTH-ORDER POLYNOMIAL INTERPOLATION

nth-Order Polynomial Interpolation

$$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

- (n+1) data points are required

$$[x_0, f(x_0)], [x_1, f(x_1)], \dots, [x_n, f(x_n)]$$



Coefficients to be evaluated:

$f[]$: finite divided difference

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

.

.

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$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

First finite divided difference

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

Second finite divided difference

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

nth finite divided difference

NTH-ORDER POLYNOMIAL INTERPOLATION

Finite Divided Difference:

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

First finite divided difference

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

Second finite divided difference

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

nth finite divided difference

recursive

Higher order differences are computed by taking differences of lower-order differences

<i>i</i>	x_i	$f(x_i)$	First	Second	Third
0	x_0	$f(x_0)$	$f[x_1, x_0]$	$f[x_2, x_1, x_0]$	$f[x_3, x_2, x_1, x_0]$
1	x_1	$f(x_1)$	$f[x_2, x_1]$	$f[x_3, x_2, x_1]$	
2	x_2	$f(x_2)$	$f[x_3, x_2]$		
3	x_3	$f(x_3)$			

NTH-ORDER POLYNOMIAL INTERPOLATION

Finite Divided Difference (Detailed Steps):

$$f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$f[x_2, x_1] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$f[x_3, x_2] = \frac{f(x_3) - f(x_2)}{x_3 - x_2}$$

$$f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

$$f[x_3, x_2, x_1] = \frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$$

$$f[x_3, x_2, x_1, x_0] = \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$$

recursive

Used to compute interpolating polynomial

Used to compute the finite divided difference at the next order

NTH-ORDER POLYNOMIAL INTERPOLATION

nth-Order Polynomial Interpolation

$$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

.

.

.

$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$



$$f_n(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] \\ + \cdots + (x - x_0)(x - x_1) \cdots (x - x_{n-1})f[x_n, x_{n-1}, \dots, x_0]$$

Newton's divided-difference interpolating polynomial

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- Data points do not need to be equally spaced
 - The abscissa values do not need to be in ascending order

LAGRANGE INTERPOLATING POLYNOMIAL

- A reformulation of the Newton polynomial without computing divided differences

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

Known value @ x_i

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

“product of”

$$L_1(x) = \frac{(x - x_0)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(\cdots)(x_1 - x_n)}$$

(n+1) data points available



$L_i(x)$ will be 1 at $x = x_i$ and 0 at all the other sample points x

$$f(x) = \frac{(x - x_1)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(\cdots)(x_0 - x_n)} f(x_0) + \frac{(x - x_0)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(\cdots)(x_1 - x_n)} f(x_1) + \dots + \frac{(x - x_0)(x - x_1)(x - x_2)(\cdots)(x - x_{n-1})}{(x_n - x_0)(x_n - x_2)(x_n - x_3)(\cdots)(x_n - x_{n-1})} f(x_n)$$

Each product $L_i(x)f(x_i)$ takes on the value of $f(x_i)$ at the sample point x_i

LAGRANGE INTERPOLATING POLYNOMIAL

Linear Lagrange Interpolating Polynomial (n = 1)

$$f(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1) \quad \text{2 data points available}$$
$$= \frac{f(x_1)(x - x_0) - f(x_0)(x - x_1)}{(x_1 - x_0)} = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} x + \frac{x_1 f(x_0) - x_0 f(x_1)}{(x_1 - x_0)}$$

2nd-order Lagrange Interpolating Polynomial (n = 3)

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1)$$
$$+ \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

NUMERICAL INTEGRATION

- Replacing a true (but unknown) or a complicated function or tabulated data with an approximating function

$$I = \int_a^b f(x) dx \cong \int_a^b f_n(x) dx$$

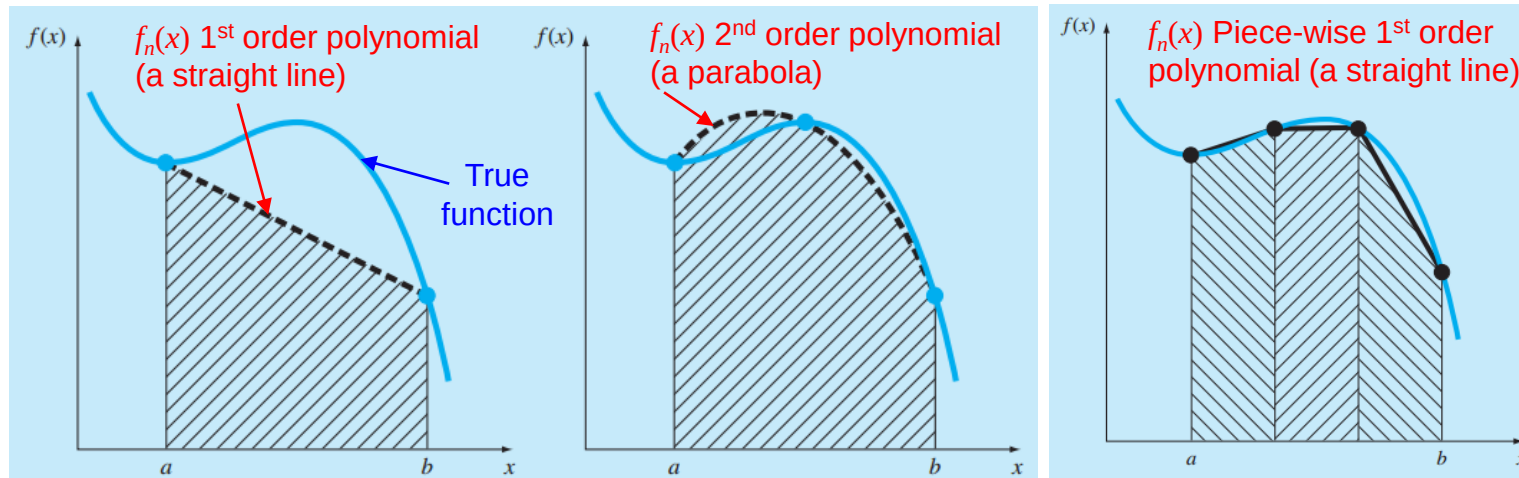
True (but unknown) or
complicated function

An approximating
function

$f_n(x)$ is in a polynomial form:

$$f_n(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1} + a_nx^n$$

n : order of the polynomial



TRAPEZOIDAL RULE

The Trapezoidal Rule:

- The easiest scheme for numerical integration
- The approximating polynomial is the first-order (a line)

$$I = \int_a^b f(x) dx \cong \int_a^b f_1(x) dx$$

- For a linear interpolation within $x = [a, b]$

$$f_1(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$$

↓ Substitute $f_1(x)$ into the integral I

$$I = \int_a^b \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right] dx$$

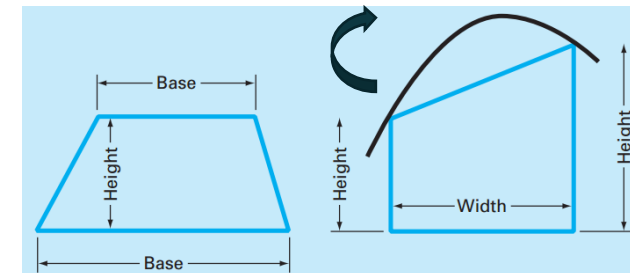
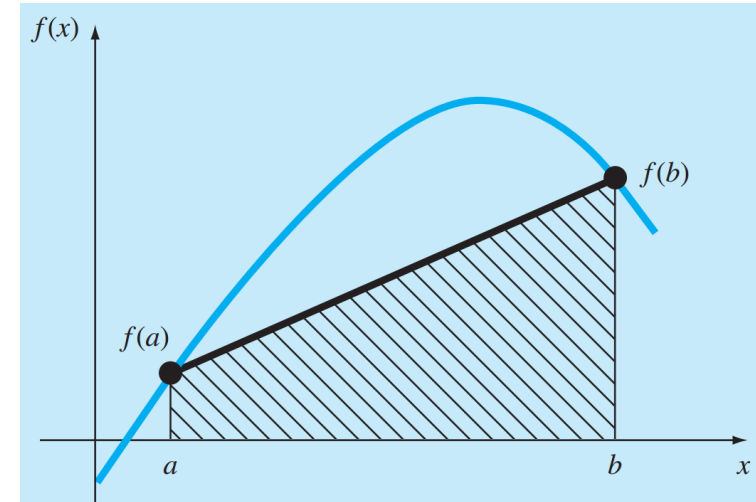
Trick: define $\tilde{x} = x - a$ and $\tilde{x}$ in the range $[0, b - a]$

$$I = (b - a) \frac{f(a) + f(b)}{2}$$

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Trapezoidal rule →

Equivalent to approximating
area of the trapezoid



Engineering
Computing

THE MULTIPLE-APPLICATION TRAPEZOIDAL RULE

- Divide the integration interval into n segments ($n+1$ base points)
- Each segment

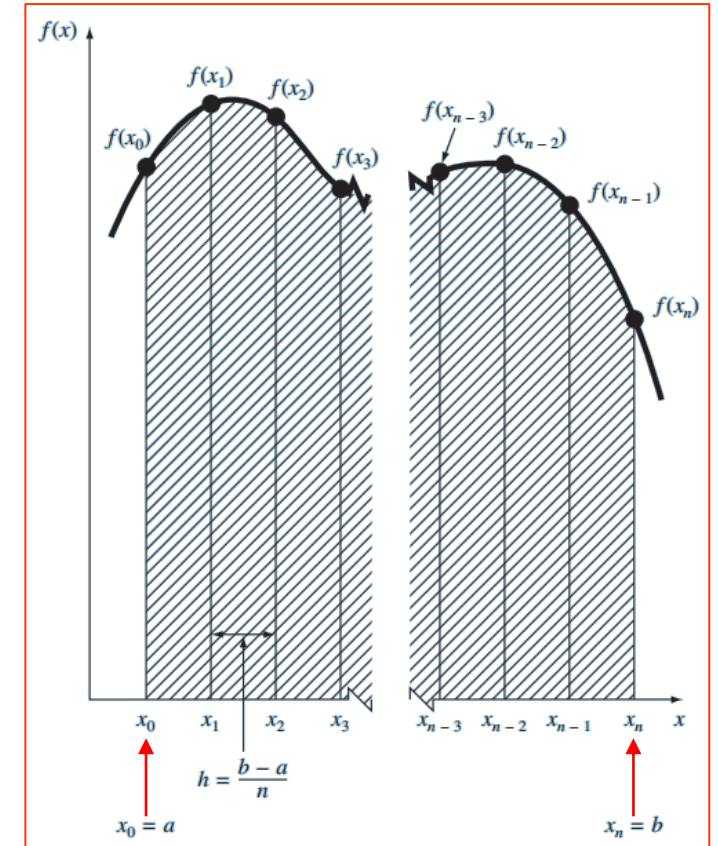
$$h = \frac{b - a}{n}$$

$$I = \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \cdots + \int_{x_{n-1}}^{x_n} f(x) dx$$

$$I = h \frac{f(x_0) + f(x_1)}{2} + h \frac{f(x_1) + f(x_2)}{2} + \cdots + h \frac{f(x_{n-1}) + f(x_n)}{2}$$

$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

$$I = \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$



SIMPSON'S 1/3 RULE

The Simpson's 1/3 Rule:

- Using a "second-order polynomial" as the approximating function

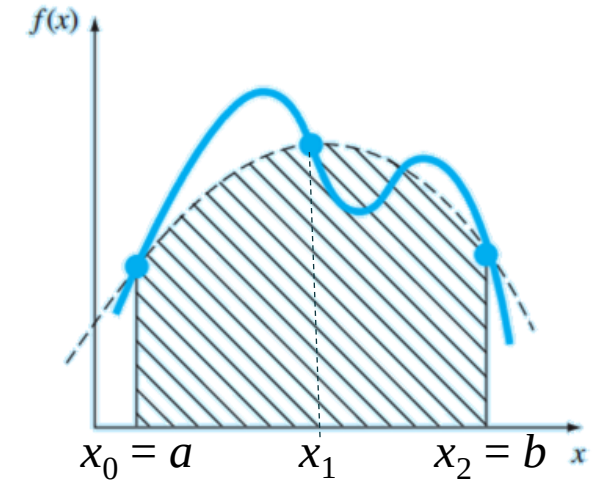
$$I \cong \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

"1/3" rule

$$h = (b-a)/2$$



$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}}_{\text{Average height}}$$



Multiple-Application Simpson's 1/3 Rule

$$I \cong 2h \frac{f(x_0) + 4f(x_1) + f(x_2)}{6} + 2h \frac{f(x_2) + 4f(x_3) + f(x_4)}{6} + \dots + 2h \frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6}$$

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{j=2,4,6}^{n-2} f(x_j) + f(x_n)}{3n}}_{\text{Average height}}$$

SIMPSON'S 3/8 RULE

The Simpson's 1/3 Rule:

- Only applicable to an even number of segments and an odd number of points

The Simpson's 3/8 Rule:

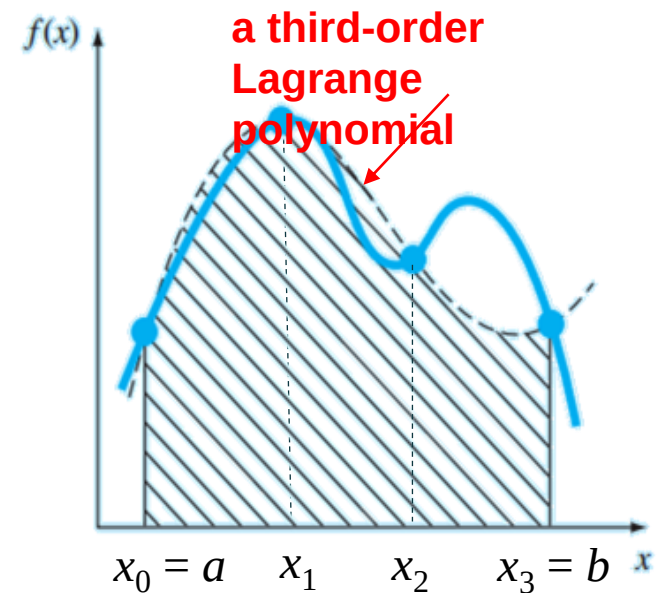
- Applicable to an odd number of segments and an even number of points

The Combination of Simpson's 1/3 and 3/8 Rule will allow evaluation of both cases

The Simpson's 3/8 Rule:

$$I = \int_a^b f(x) dx \cong \int_a^b f_3(x) dx$$

- A 3rd-order polynomial for $f_3(x)$ within $x = [a \ b]$



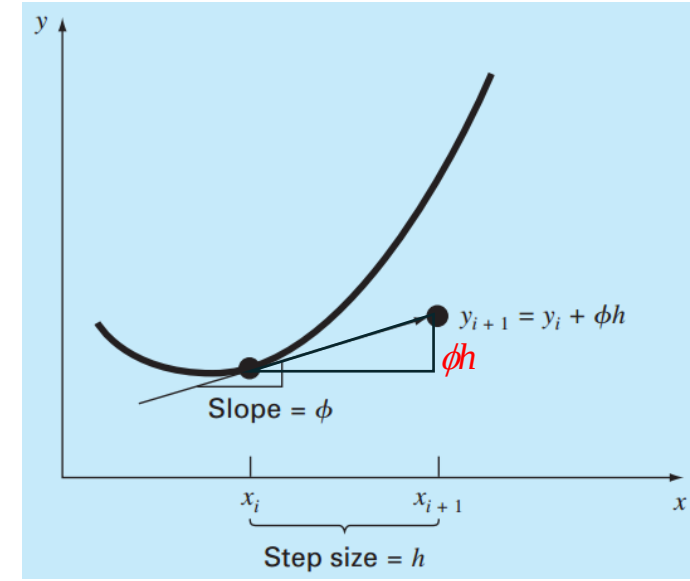
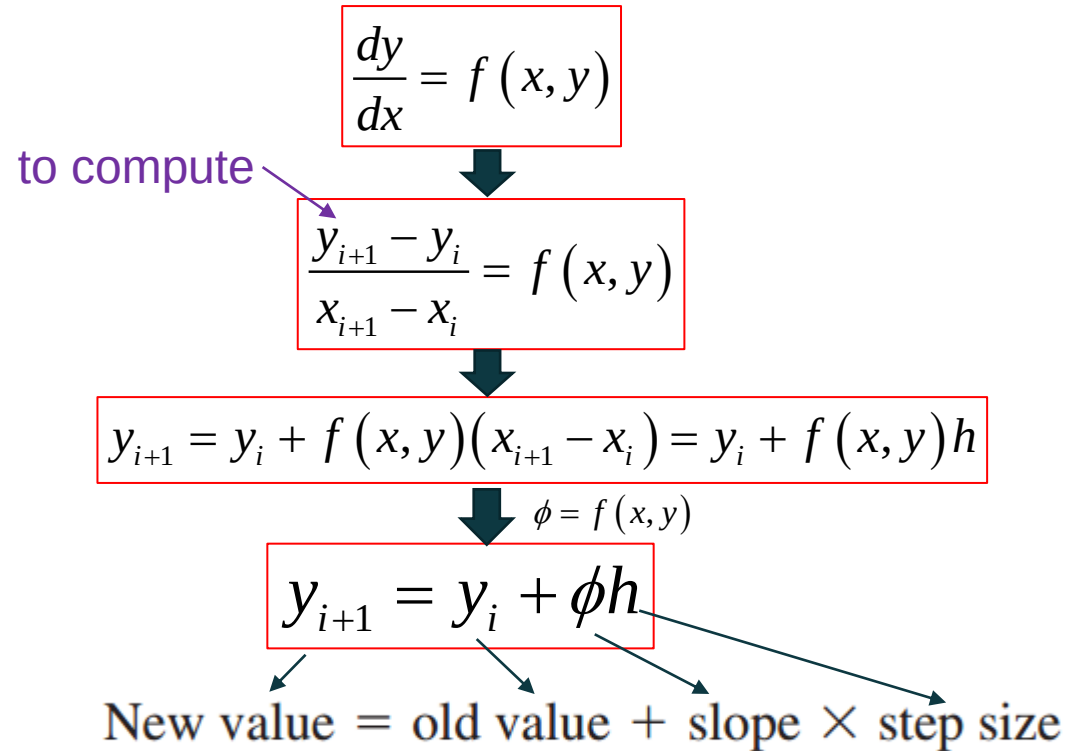
$$I \cong \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

"3/8" rule

$$h = (b-a)/3$$

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}}_{\text{Average height}}$$

ORDINARY DIFFERENTIAL EQUATIONS : RUNGE-KUTTA METHOD



ϕ :

- The slope estimate
- Used to extrapolate from an old value y_i to a new value y_{i+1} over a distance h
- The different methods to estimate the slope ϕ lead to different numerical solution methods
- All generally called "*Runge-Kutta*" methods

EULER'S METHOD

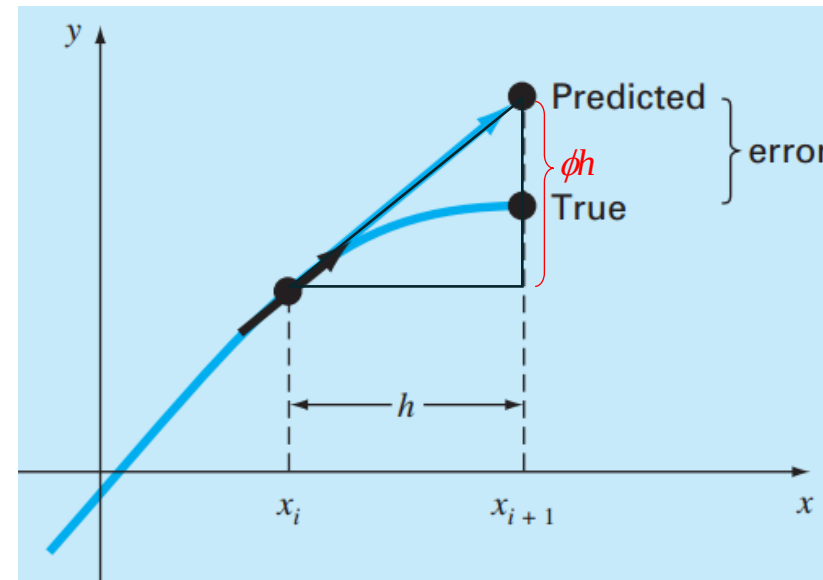
$$y_{i+1} = y_i + \phi h$$

New value = old value + slope $\times$ step size

$$\phi = f(x_i, y_i)$$

ϕ direct estimate of the slope at x_i

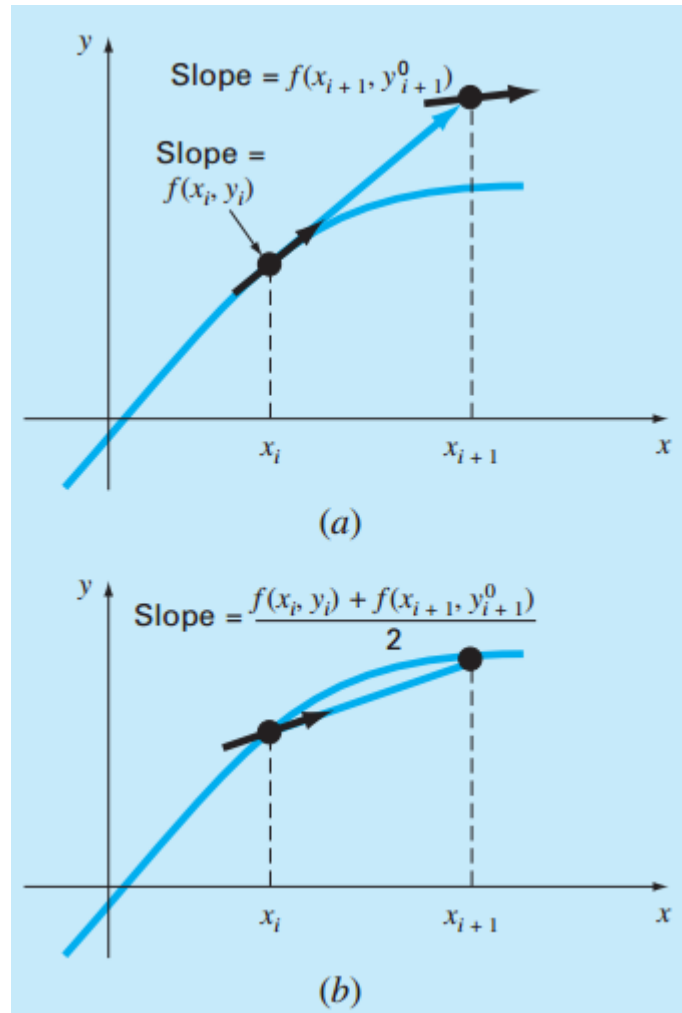
$$y_{i+1} = y_i + f(x_i, y_i)h$$



- Prediction based on the first-order Taylor expansion
- A new value of y is predicted using the slope (the first derivative at the original value of x) to extrapolate linearly over the step size h

HEUN'S METHOD

Graphic Interpretation



Predictor

$$y_{i+1}^0 = y_i + f(x_i, y_i)h \quad \text{Euler's Method}$$

$$y'_{i+1} = f(x_{i+1}, y_{i+1}^0) \quad \text{Slope estimate at end points } y'_{i+1}$$

Corrector

$$y_{i+1} = y_i + \left[\frac{f(x_i, y_i) + f(x_{i+1}, y_{i+1}^0)}{2} \right] h$$